NETWORK LAYER

Introduction • Need • Services Provided • Design Issues

DETAILED COMPUTER NETWORKS NOTES

Concept explanations • Colour-coded diagrams • Comparison tables • Worked examples • Revision questions

Learning outcomes
By the end of these notes, you should be able to explain the role of the Network Layer, why it is required, the services it offers to the Transport Layer, and the major design issues involved in delivering packets across interconnected networks.

Big idea: The Network Layer makes communication possible between devices on different networks. It uses logical addressing and routing to move packets from a source host toward a destination host.

OSI model position

Layer	Example / responsibility
7. Application	HTTP, DNS, email applications
6. Presentation	Data representation, encryption formats
5. Session	Dialog/session management (conceptual OSI role)
4. Transport	TCP / UDP; process-to-process delivery
3. Network	IP addressing, routing, packet forwarding
2. Data Link	Frames, MAC addresses, local-link delivery
1. Physical	Signals, bits, cables, radio

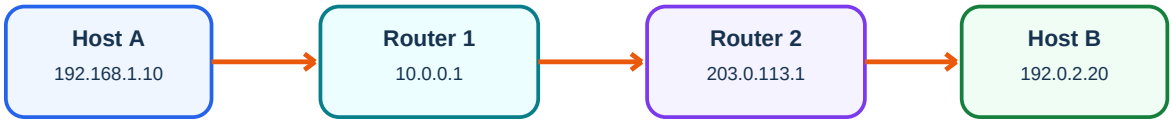
Remember: switches usually forward frames within a LAN using Layer 2 information; routers forward IP packets between networks using Layer 3 information.

1. Introduction to the Network Layer

The **Network Layer** is Layer 3 of the OSI reference model. Its main job is to deliver packets from a source host to a destination host, even when the two hosts are located on different networks. It provides logical addressing and chooses or follows routes through routers and interconnected links.

The best-known Network Layer protocol is **Internet Protocol (IP)**, including IPv4 and IPv6. Routing protocols such as OSPF and BGP help routers learn paths; they support routing decisions but are not themselves the packet format used for ordinary IP data delivery.

Logical end-to-end communication across multiple networks



Routers examine the destination IP address and forward packets toward the destination network.

Packet delivery: a simple example

Suppose a student's laptop has IP address **192.168.1.10** and opens a server at **192.0.2.20**. If the destination is outside the laptop's local subnet, the laptop sends the packet to its default gateway. Routers inspect the destination IP address, consult their forwarding tables, and pass the packet to the next hop. The packet may cross several networks before reaching the destination.

> **Key distinction: the IP address identifies the logical source and destination across networks. The link-layer address (such as an Ethernet MAC address) is used to deliver a frame over a particular local link.**

2. Why Do We Need the Network Layer?

A single LAN can deliver frames locally, but the modern internet contains millions of separate networks using different technologies. The Network Layer provides a common way to address destinations and forward packets across those networks.

Need	Explanation	Example
Inter-network communication	Connects devices that are not on the same LAN.	A college laptop reaches a website hosted in another country.
Logical addressing	Uses IP addresses independent of a device's physical network interface.	A server can be reached by its IP address even across many routers.
Path selection (routing)	Determines suitable paths through a network of routers.	A router chooses a next hop toward a destination prefix.
Packet forwarding	Moves each packet from an incoming interface to an outgoing interface.	A router forwards a packet from its WAN link toward the campus LAN.
Heterogeneous networks	Allows communication across different link technologies.	Wi-Fi at home, fibre in an ISP network, Ethernet in a data centre.
Scalability	Hierarchical addressing and route aggregation help large networks operate.	A provider advertises one address prefix representing many customer addresses.
Handling failures	Routing can adapt when a link or router becomes unavailable.	Traffic takes an alternate route after a link failure, if one exists.

Without the Network Layer

Each network technology would need its own unrelated mechanism for reaching every remote network. Applications would have to understand many different link technologies and paths. IP and routing provide a common layer that hides much of this complexity from applications.

3. Services Provided by the Network Layer

The Network Layer offers a service interface to the Transport Layer. The exact guarantees depend on the network architecture. In the Internet, IP generally provides a **connectionless, best-effort datagram service**: it attempts to deliver packets but does not guarantee delivery, order, or a fixed delay.

Service / property	Meaning	Internet IP example
Logical addressing	Identifies source and destination hosts/interfaces using network-layer addresses.	IPv4 or IPv6 source and destination fields.
Routing	Computes or learns paths through routers.	Routers learn routes using static configuration or routing protocols.
Forwarding	Sends an arriving packet to the appropriate next hop.	Lookup destination prefix in a forwarding table.
Internetworking	Carries packets across multiple interconnected networks.	Packet crosses home router, ISP routers, and destination network.
Fragmentation / MTU handling	Deals with differences in maximum packet size across links; mechanisms differ by IP version.	IPv4 routers may fragment in some circumstances; IPv6 routers do not fragment packets in transit, so endpoints use Path MTU Discovery.
Congestion-related support	Networks may detect or signal congestion, but basic IP does not promise congestion-free delivery.	Queueing delay, packet drops, or ECN marking may occur.
Quality of Service (QoS) support	Fields and mechanisms can help classify traffic for differentiated handling.	DSCP markings may influence treatment when the network is configured to honor them.

Connectionless service vs connection-oriented service

Connectionless (datagram)	Connection-oriented (virtual-circuit model)
Each packet is handled independently and carries a destination address.	A logical path or connection is established before data transfer.
Packets from the same application flow may follow different routes.	Packets commonly follow the established virtual circuit.
Routers need not maintain per-flow connection state for ordinary IP forwarding.	Network devices may maintain state for the virtual circuit.
Example: the Internet Protocol (IP).	Classic textbook example: a virtual-circuit network such as X.25 or ATM.

Important exam point: IP is best-effort. Reliability, retransmission, and in-order delivery—when required—are typically handled by higher-layer protocols such as TCP, not guaranteed by IP itself.

4. Network Layer Design Issues

Network-layer designers must decide how packets are addressed, routed, forwarded, queued, and handled when the network is overloaded or links have different characteristics. The major issues are described below.

4.1 Store-and-Forward Packet Switching

In a store-and-forward packet-switched network, a router receives a packet (or enough of it for the forwarding process), checks relevant information, and forwards it toward the next hop. Each hop introduces processing and queueing delay; serialization and propagation also contribute to total delay.

> **Total end-to-end delay is influenced by processing delay + queueing delay + transmission delay + propagation delay at each relevant hop.**

4.2 Services to the Transport Layer

The network layer must define what the Transport Layer can expect. A design may offer a connectionless datagram service or a connection-oriented virtual-circuit service. The choice affects router state, failure handling, packet ordering, and service guarantees.

4.3 Connectionless vs Connection-Oriented Network Design

In a datagram design, each packet is routed independently using its destination address. In a virtual-circuit design, a path is set up first and packets carry a short circuit identifier. Neither design is universally best; each makes different trade-offs in flexibility, state, and control.

4.4 Routing and Forwarding

Routing is the process of selecting paths and building routing information. Forwarding is the per-packet action of selecting the outgoing interface or next hop. A router may use routing protocols to populate its routing table, then install suitable entries in its forwarding table.

Term	Question answered	Example
Routing	Which path should traffic use?	A routing protocol selects a route to 203.0.113.0/24.
Forwarding	Where should this packet go next?	A packet for 203.0.113.25 leaves through interface 2.

4.5 Addressing and Internetworking

IP addresses must be structured so routers can determine which network a destination belongs to. Subnet masks and prefixes (such as **192.168.1.0/24**) identify network portions. Address planning, subnetting, route aggregation, and private/public address use all affect how large networks scale.

4.6 Congestion Control and Resource Allocation

Congestion occurs when packets arrive at a router faster than its outgoing links can send them. Queues grow, delay increases, and buffers may overflow, causing packet loss. Network design may use queue management, traffic engineering, admission policies, or explicit congestion signals. End hosts and transport protocols also play important roles.

4.7 Quality of Service (QoS)

Different traffic types have different needs. Voice and interactive video are sensitive to delay and jitter; bulk file transfer often prioritizes throughput and accuracy. QoS mechanisms can classify, mark, schedule, or police traffic, but a QoS field alone does not guarantee performance across the entire path.

4.8 Internetworking Across Different MTUs

Different link technologies support different maximum transmission units (MTUs). A packet that is too large for a link may need fragmentation or may be dropped with a message indicating that a smaller packet is required. IPv4 and IPv6 handle router fragmentation differently, so endpoint MTU discovery is important.

4.9 Error Handling and Diagnostics

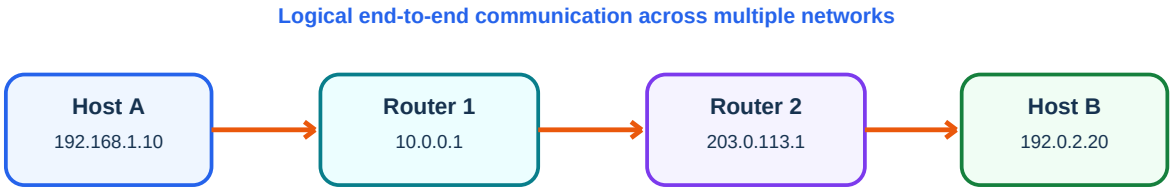
The network layer needs ways to report certain delivery problems and support diagnostics. ICMP and ICMPv6 carry control and error messages such as destination unreachable or time exceeded. Tools such as ping and traceroute use these mechanisms in different ways, although firewalls and network policies may limit responses.

4.10 Security

Network-layer design must account for spoofed addresses, unauthorized traffic, route manipulation, denial-of-service attacks, and exposure of traffic metadata. Filtering, secure routing practices, monitoring, and IPsec where appropriate can help, but no single mechanism solves every security problem.

5. Diagrams and Worked Examples

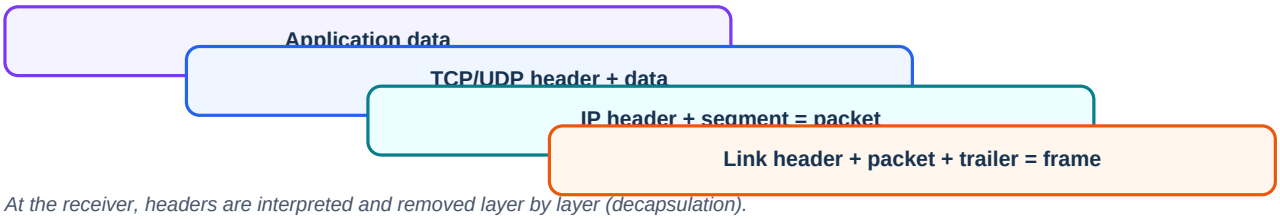
Diagram A: forwarding across networks



Routers examine the destination IP address and forward packets toward the destination network.

A router does not usually send an Ethernet frame unchanged from end to end. At each routed hop, the incoming link-layer frame is removed, the IP packet is examined, and a new link-layer frame is built for the next link. The source and destination IP addresses generally remain the same end-to-end, except when functions such as NAT modify them; link-layer addresses change at each hop.

Diagram B: encapsulation



At the receiver, headers are interpreted and removed layer by layer (decapsulation).

Data from an application is passed down the protocol stack. Transport adds its header to form a segment or datagram; IP adds a network-layer header to form a packet; the link layer wraps the packet in a frame. The receiver processes the layers in reverse order.

Worked Example 1: deciding whether to use the gateway

A laptop has IP address **192.168.10.25/24** and wants to reach **192.168.10.80**. With a /24 prefix, both addresses are in the **192.168.10.0/24** network, so the laptop can attempt local delivery using the link layer (for example, ARP in IPv4 Ethernet). Now suppose the destination is **8.8.8.8**. That address is outside the local /24 network, so the laptop sends the packet to its configured default gateway.

> **Exam tip: hosts normally compare the destination IP with their own IP and subnet mask/prefix to decide whether the destination is local or must be reached through a router.**

Worked Example 2: routing table lookup

Destination prefix	Next hop / interface
192.168.1.0/24	Directly connected — interface LAN
10.20.0.0/16	Router A — interface 1
0.0.0.0/0	Default gateway — interface WAN

Suppose a packet is destined for **10.20.5.9**. The router looks for the most specific matching prefix. The entry **10.20.0.0/16** matches, so Router A is selected. If no more specific entry matched, the default route 0.0.0.0/0 could be used. This principle is called **longest-prefix match**.

Worked Example 3: congestion

Imagine a router receives traffic at 1.2 Gb/s but its outgoing link can transmit only 1 Gb/s. If this imbalance continues, the queue grows. The router may add delay and eventually drop packets when its buffer fills. Possible responses include reducing offered load, choosing a different route, improving capacity, or using congestion signalling and traffic-management mechanisms.

6. Summary for Quick Revision

Question	Short answer
What is the Network Layer?	OSI Layer 3; delivers packets across interconnected networks.
Main protocol?	IP (IPv4 and IPv6).
Main address type?	Logical IP address.
What is routing?	Selecting paths and learning route information.
What is forwarding?	Sending a packet to the next hop/outgoing interface.
What service does IP provide?	Connectionless, best-effort datagram delivery.
Does IP guarantee delivery or order?	No. Higher layers may provide reliability and ordering.
What is congestion?	Offered traffic exceeds available forwarding capacity.
What is MTU?	Maximum size of a packet/frame payload supported on a link, depending on the layer and definition.
What is ICMP used for?	Control/error reporting and diagnostics.

7. Practice Questions

1. Define the Network Layer and state its major responsibilities.
2. Explain at least five reasons why the Network Layer is needed.
3. Differentiate between routing and forwarding with an example.
4. Compare connectionless datagram service and connection-oriented virtual-circuit service.
5. What does best-effort delivery mean? Name two guarantees IP does not provide.
6. Explain congestion and describe two ways to respond to it.
7. What is the purpose of logical addressing and subnet prefixes?
8. Explain how a host decides whether to send a packet directly or via its default gateway.
9. What is MTU, and why can different MTUs create a packet-delivery problem?
10. Describe encapsulation as data moves down the protocol stack.

8. Answer Hints

1. Layer 3; logical addressing, routing and forwarding. 2. Inter-network communication, logical addressing, routing, forwarding, heterogeneous link support, scalability, failure recovery. 3. Routing selects a path; forwarding moves a specific packet to the next hop. 4. Datagram packets are handled independently; virtual circuits establish a path/state first. 5. IP tries to deliver but does not guarantee delivery, order, or fixed delay. 6. Queue growth due to overload; manage queues, signal congestion, reroute, or add capacity. 7. Prefixes identify destination networks and enable aggregation. 8. Compare destination to local subnet; local destinations use link-layer delivery, remote destinations use the gateway. 9. MTU is a size limit; oversized packets may require handling or be dropped. 10. Transport header, IP header, then link-layer framing.

> **Final takeaway**
> **The Network Layer connects networks into an internetwork. Its core functions are logical addressing, routing, and forwarding. Its design must balance scalability, performance, reliability expectations, congestion, differing link characteristics, diagnostics, and security.**

Suggested references: standard Computer Networks textbooks (e.g., Andrew S. Tanenbaum and David J. Wetherall, *Computer Networks*) and the Internet Protocol specifications published by the IETF (RFC 791 for IPv4, RFC 8200 for IPv6). These notes are an educational summary; implementations and protocol details may vary.